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Full body posterior coil for a magnetic resonance imaging system

Abstract

A magnetic resonance imaging (MRI) system includes an MRI scanner having a bore. The MRI system also includes a table including a cradle and a posterior radio frequency (RF) receiving coil array assembly integrated on a top surface of the cradle, wherein the table is configured to move a subject to be imaged disposed on the RF receiving coil array assembly into and out of the bore of the MRI scanner, and wherein the posterior RF receiving coil array assembly comprises a plurality of sections, each section of the plurality of sections includes an RF coil including a plurality of loops, and at least one section of the plurality of sections is configured to be manipulated into a bent position and to remain in the bent position without extraneous support.

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Primary Examiner: Raymond; Keith M*Assistant Examiner:* Jasani; Ashish S*Attorney, Agent or Firm:* Fletcher Yoder, P.C.**Background/Summary****BACKGROUND**

(1) The subject matter disclosed herein relates to medical imaging and, more particularly, to a full body coil for a magnetic resonance imaging (MRI) system.

(2) Non-invasive imaging technologies allow images of the internal structures or features of a patient/object to be obtained without performing an invasive procedure on the patient/object. In particular, such non-invasive imaging technologies rely on various physical principles (such as the

differential transmission of X-rays through a target volume, the reflection of acoustic waves within the volume, the paramagnetic properties of different tissues and materials within the volume, the breakdown of targeted radionuclides within the body, and so forth) to acquire data and to construct images or otherwise represent the observed internal features of the patient/object.

(3) | During MRI, when a substance such as human tissue is subjected to a uniform magnetic field (polarizing field $B_{sub.0}$), the individual magnetic moments of the spins in the tissue attempt to align with this polarizing field, but precess about it in random order at their characteristic Larmor frequency. If the substance, or tissue, is subjected to a magnetic field (excitation field $B_{sub.1}$) which is in the x-y plane and which is near the Larmor frequency, the net aligned moment, or “longitudinal magnetization”, $M_{sub.z}$, may be rotated, or “tipped”, into the x-y plane to produce a net transverse magnetic moment, $M_{sub.t}$. A signal is emitted by the excited spins after the excitation signal $B_{sub.1}$ is terminated and this signal may be received and processed to form an image.

(4) When utilizing these signals to produce images, magnetic field gradients ($G_{sub.x}$, $G_{sub.y}$, and $G_{sub.z}$) are employed. Typically, the region to be imaged is scanned by a sequence of measurement cycles in which these gradient fields vary according to the particular localization method being used. The resulting set of received nuclear magnetic resonance (NMR) signals are digitized and processed to reconstruct the image using one of many well-known reconstruction techniques.

(5) In MRI, the workflow around removing and/or installing surface coils (e.g., for imaging particular areas of a patient such as a head, foot, leg, etc.) and the numerous patient positioning pads between patients can be cumbersome and time consuming depending on the type of scan and patient body types. The current rigid surface coils also do not perfectly fit all patients, thus, requiring shimming pads to get as close as possible. The closer the coil loops are to the selected anatomy, the better.

BRIEF DESCRIPTION

(6) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(7) In one embodiment, a magnetic resonance imaging (MRI) system is provided. The MRI system includes an MRI scanner having a bore. The MRI system also includes a table including a cradle and a posterior radio frequency (RF) receiving coil array assembly integrated on a top surface of the cradle, wherein the table is configured to move a subject to be imaged disposed on the RF receiving coil array assembly into and out of the bore of the MRI scanner, and wherein the posterior RF receiving coil array assembly includes a plurality of sections, each section of the plurality of sections includes an RF coil including a plurality of loops, and at least one section of the plurality of sections is configured to be manipulated into a bent position and to remain in the bent position without extraneous support.

(8) In another embodiment, a posterior radio frequency (RF) receiving coil array assembly for a magnetic resonance imaging (MRI) system is provided. The posterior RF receiving coil array assembly includes a plurality of sections, wherein each section of the plurality of sections includes an RF coil including a plurality of loops, and at least one section of the plurality of sections is configured to be manipulated into a bent position and to remain in the bent position without extraneous support, wherein the posterior RF receiving coil array assembly is configured to be integrated on a top surface of a cradle of a table configured to move a subject to be imaged disposed on the RF receiving coil array assembly into and out of a bore of an MRI scanner of the MRI system.

(9) In a further embodiment, a posterior RF receiving coil array assembly for a magnetic resonance imaging (MRI) system is provided. The posterior RF receiving coil array assembly a plurality of

padded sections, each padded section of the plurality of padded sections includes an RF coil including a plurality of loops, and multiple padded sections of the plurality of padded sections include a mechanically resistive structure configured both to be manipulated so that a respective padded section is in a bent position and to keep the respective padded section in the bent position without extraneous support, wherein the posterior RF receiving coil array assembly is configured to be integrated on a top surface of a cradle of a table configured to move a subject to be imaged disposed on the RF receiving coil array assembly into and out of a bore of an MRI scanner of the MRI system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects, and advantages of the present subject matter will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

- (2) FIG. 1 illustrates an embodiment of a magnetic resonance imaging (MRI) system suitable for use with the disclosed technique;
- (3) FIG. 2 is a schematic diagram of an RF coil array section and its topology, in accordance with aspects of the present disclosure;
- (4) FIG. 3 is a schematic diagram of an RF coil array section and its components, in accordance with aspects of the present disclosure;
- (5) FIG. 4 is a schematic diagram of an RF coil array section and its components (e.g., having padding incorporated in an enclosure), in accordance with aspects of the present disclosure;
- (6) FIG. 5 is a schematic diagram of an RF coil array section and its components (e.g., lacking a mechanically resistive structure), in accordance with aspects of the present disclosure;
- (7) FIG. 6 is a schematic diagram of an RF coil array and its components (e.g., lacking a mechanically resistive structure and having padding incorporated in an enclosure), in accordance with aspects of the present disclosure;
- (8) FIG. 7 is a schematic diagram of a posterior RF receiving coil array assembly disposed on a cradle of a table of the MRI system in FIG. 1, in accordance with aspects of the present disclosure;
- (9) FIG. 8 is a schematic diagram of an end view of a posterior RF receiving coil array assembly disposed within a bore of an MRI system, in accordance with aspects of the present disclosure;
- (10) FIG. 9 is a schematic diagram of RF array coil sections of a posterior RF receiving coil array assembly disposed about a portion of a patient, in accordance with aspects of the present disclosure;
- (11) FIG. 10 is a schematic diagram of a top view of a posterior RF receiving coil array assembly being utilized with a pediatric patient, in accordance with aspects of the present disclosure;
- (12) FIG. 11 is a schematic diagram of RF array coil sections of a posterior RF receiving coil array assembly disposed about a portion of a patient, in accordance with aspects of the present disclosure;
- (13) FIG. 12 is a schematic diagram of an RF coil array section of a posterior RF receiving coil array assembly being positioned to lift a portion of a patient (e.g., head), in accordance with aspects of the present disclosure; and
- (14) FIG. 13 is a schematic diagram of an RF coil array section of a posterior RF receiving coil array assembly being positioned to lift a portion of a patient (e.g., legs and feet), in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

- (15) One or more specific embodiments will be described below. In an effort to provide a concise

description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(16) When introducing elements of various embodiments of the present subject matter, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical values, ranges, and percentages are within the scope of the disclosed embodiments.

(17) While aspects of the following discussion are provided in the context of medical imaging, it should be appreciated that the disclosed techniques are not limited to such medical contexts. Indeed, the provision of examples and explanations in such a medical context is only to facilitate explanation by providing instances of real-world implementations and applications. However, the disclosed techniques may also be utilized in other contexts, such as image reconstruction for non-destructive inspection of manufactured parts or goods (i.e., quality control or quality review applications), and/or the non-invasive inspection of packages, boxes, luggage, and so forth (i.e., security or screening applications). In general, the disclosed techniques may be useful in any imaging or screening context or image processing or photography field where a set or type of acquired data undergoes a reconstruction process to generate an image or volume.

(18) The present disclosure provides for a full body coil for an MRI system. In particular, a posterior radio frequency (RF) receiving coil array assembly is provided that is integrated on a top surface of a cradle of a table that moves a patient (e.g., subject) into and out of a bore of an MRI scanner of the MRI system. The posterior RF receiving coil array assembly is formed by multiple sections (e.g. padded sections). Each section includes an RF coil having multiple loops (e.g., elements or channels). One or more sections are configured to be manipulated into a bent position and to remain in the bent position without extraneous support. In particular, the one or more sections are configured to be bent about the patient's anatomy from a posterior surface toward an anterior surface (and in some cases disposed about a portion of the anterior surface). Each of the sections is configured to be bent or manipulated include a mechanically resistive structure (e.g., disposed within a padded enclosure) to keep the sections in the bent position without extraneous support (e.g., from shims). The posterior RF receiving coil array assembly extends across an entirety of the top surface of the cradle. In certain embodiments, the posterior RF receiving coil array assembly can be utilized on both a posterior surface and an anterior surface of a pediatric patient without utilizing any additional RF receiving coil array assembly during an imaging scan of the pediatric patient by the MRI system.

(19) The disclosed embodiments enable the posterior RF receiving coil array to be disposed closer to the anatomy of the patient being imaged. The disclosed embodiments enable for better in-table patient positioning and remove the need for comfort tilt accessories. In addition, the disclosed embodiments alleviate issues of carrying around large and somewhat clumsy surface coils for quicker workflow. The disclosed embodiments further provide better contour fitting and comfort to different patient types which boosts image quality. The disclosed embodiments reduce trips to coils closet and overall setup time for the patients while also providing a closer loop-to-anatomy distance to optimize image quality. Even further, by improving workflow with less trips and providing more integrated capability all at the table, the throughput of scanning patients is increased.

(20) With the preceding in mind, FIG. 1 a magnetic resonance imaging (MRI) system **100** is

illustrated schematically as including a scanner **102**, scanner control circuitry **104**, and system control circuitry **106**. According to the embodiments described herein, the MRI system **100** is generally configured to perform MR imaging.

(21) System **100** additionally includes remote access and storage systems or devices such as picture archiving and communication systems (PACS) **108**, or other devices such as teleradiology equipment so that data acquired by the system **100** may be accessed on- or off-site. In this way, MR data may be acquired, followed by on- or off-site processing and evaluation. While the MRI system **100** may include any suitable scanner or detector, in the illustrated embodiment, the system **100** includes a full body scanner **102** having a housing **120** through which a bore **122** is formed. A table **124** is moveable into the bore **122** to permit a patient **126** (e.g., subject) to be positioned therein for imaging selected anatomy within the patient **126**. The portion of the table **124** that the patient is **126** disposed on and that is moved into and out of the bore **122** is a cradle **127**.

(22) Scanner **102** includes a series of associated coils for producing controlled magnetic fields for exciting the gyromagnetic material within the anatomy of the patient being imaged. Specifically, a primary magnet coil **128** is provided for generating a primary magnetic field, B.sub.0, which is generally aligned with the bore **122**. A series of gradient coils **130**, **132**, and **134** permit controlled magnetic gradient fields to be generated for positional encoding of certain gyromagnetic nuclei within the patient **126** during examination sequences. A radio frequency (RF) coil **136** (e.g., RF transmit coil) is configured to generate radio frequency pulses for exciting the certain gyromagnetic nuclei within the patient. In addition to the coils that may be local to the scanner **102**, the system **100** also includes a set of receiving coils or RF receiving coils **138** (e.g., an array of coils) configured for placement proximal (e.g., against) to the patient **126**. As an example, the receiving coils **138** can include cervical/thoracic/lumbar (CTL) coils, head coils, single-sided spine coils, and so forth. Generally, the receiving coils **138** are placed close to or on top of the patient **126** so as to receive the weak RF signals (weak relative to the transmitted pulses generated by the scanner coils) that are generated by certain gyromagnetic nuclei within the patient **126** as they return to their relaxed state.

(23) The various coils of system **100** are controlled by external circuitry to generate the desired field and pulses, and to read emissions from the gyromagnetic material in a controlled manner. In the illustrated embodiment, a main power supply **140** provides power to the primary field coil **128** to generate the primary magnetic field, B.sub.0. A power input (e.g., power from a utility or grid), a power distribution unit (PDU), a power supply (PS), and a driver circuit **150** may together provide power to pulse the gradient field coils **130**, **132**, and **134**. The driver circuit **150** may include amplification and control circuitry for supplying current to the coils as defined by digitized pulse sequences output by the scanner control circuitry **104**.

(24) Another control circuit **152** is provided for regulating operation of the RF coil **136**. Circuit **152** includes a switching device for alternating between the active and inactive modes of operation, wherein the RF coil **136** transmits and does not transmit signals, respectively. Circuit **152** also includes amplification circuitry configured to generate the RF pulses. Similarly, the receiving coils **138** are connected to switch **154**, which is capable of switching the receiving coils **138** between receiving and non-receiving modes. Thus, the receiving coils **138** resonate with the RF signals produced by relaxing gyromagnetic nuclei from within the patient **126** while in the receiving mode, and they do not resonate with RF energy from the transmitting coils (i.e., coil **136**) so as to prevent undesirable operation while in the non-receiving mode. Additionally, a receiving circuit **156** is configured to receive the data detected by the receiving coils **138** and may include one or more multiplexing and/or amplification circuits.

(25) It should be noted that while the scanner **102** and the control/amplification circuitry described above are illustrated as being coupled by a single line, many such lines may be present in an actual instantiation. For example, separate lines may be used for control, data communication, power transmission, and so on. Further, suitable hardware may be disposed along each type of line for the

proper handling of the data and current/voltage. Indeed, various filters, digitizers, and processors may be disposed between the scanner and either or both of the scanner and system control circuitry **104, 106**.

(26) As illustrated, scanner control circuitry **104** includes an interface circuit **158**, which outputs signals for driving the gradient field coils and the RF coil and for receiving the data representative of the magnetic resonance signals produced in examination sequences. The interface circuit **158** is coupled to a control and analysis circuit **160**. The control and analysis circuit **160** executes the commands for driving the circuit **150** and circuit **152** based on defined protocols selected via system control circuit **106**.

(27) Control and analysis circuit **160** also serves to receive the magnetic resonance signals and performs subsequent processing before transmitting the data to system control circuit **106**. Scanner control circuit **104** also includes one or more memory circuits **162**, which store configuration parameters, pulse sequence descriptions, examination results, and so forth, during operation.

(28) Interface circuit **164** is coupled to the control and analysis circuit **160** for exchanging data between scanner control circuitry **104** and system control circuitry **106**. In certain embodiments, the control and analysis circuit **160**, while illustrated as a single unit, may include one or more hardware devices. The system control circuit **106** includes an interface circuit **166**, which receives data from the scanner control circuitry **104** and transmits data and commands back to the scanner control circuitry **104**. The control and analysis circuit **168** may include a CPU in a multi-purpose or application specific computer or workstation. Control and analysis circuit **168** is coupled to a memory circuit **170** to store programming code for operation of the MRI system **100** and to store the processed image data for later reconstruction, display and transmission. The programming code may execute one or more algorithms that, when executed by a processor, are configured to perform reconstruction of acquired data as described below. In certain embodiments, image reconstruction may occur on a separate computing device having processing circuitry and memory circuitry.

(29) An additional interface circuit **172** may be provided for exchanging image data, configuration parameters, and so forth with external system components such as remote access and storage devices **108**. Finally, the system control and analysis circuit **168** may be communicatively coupled to various peripheral devices for facilitating operator interface and for producing hard copies of the reconstructed images. In the illustrated embodiment, these peripherals include a printer **174**, a monitor **176**, and user interface **178** including devices such as a keyboard, a mouse, a touchscreen (e.g., integrated with the monitor **176**), and so forth.

(30) FIG. 2 is a schematic diagram of an RF coil array section **180** (e.g., RF receiving coil array section) and its topology. As described in greater detail below, the RF coil array section **180** is one section **180** of a plurality of sections **180** forming a posterior RF receiving coil array assembly integrated on a top surface of a cradle. The shape and dimensions of each section may vary from that depicted in FIG. 2. The RF coil array section **180** may be utilized in an MRI system (e.g., MRI system **100** in FIG. 1). The RF coil array section **180** includes an RF coil **182** (e.g., flexible RF coil) having a plurality of loops **184** (e.g., elements or channels) with each loop **184**.

(31) As depicted in FIG. 2, the plurality of loops **184** includes 8 loops (although the number of loops **184** may vary). Each loop **184** is coupled to an electronics unit **185** coupled to a coil-interfacing cable **186**. The coil-interfacing cables **186** of each of the loops **184** is coupled to a balun **187** (e.g., integrated balun cable harness). Each electronics unit **185** may include various components (e.g., a decoupling circuit, an impedance inverter circuit, and a pre-amplifier). The balun **187** may act as an RF trap. The balun **187** is coupled (via a cable **188**) to a P connector **190** (e.g., port connector) that enables the RF coil array **180** to be coupled to the interface of the MRI system that couples imaging components to processing components. In certain embodiments, the coil-interfacing cables **186** for the loops **184** of RF coils **182** of multiple sections forming the posterior RF receiving coil array assembly may be coupled to a common balun **187** coupled to the P connector **190** or coupled to separate baluns **187** coupled to a common P connector **190**.

(32) The loops **184** are disposed within an enclosure **191**. In certain embodiments, the enclosure may be flexible. As depicted in FIG. 2, each loop **184** partially overlaps with at least one adjacent loop **184**. In certain embodiments, one or more of the RF coils **182** may be designed utilizing AIR™ coil technology from General Electric Healthcare. This enables the RF coil **182** to be lightweight and flexible.

(33) FIGS. 3 and 4 are schematic diagrams of the RF coil array section **180** and its components for some of the sections forming a posterior RF receiving coil array assembly (e.g., posterior RF receiving coil array assembly **196** in FIG. 7). Each RF coil array section **180** includes the enclosure **191**. The RF coil **182** is disposed within the enclosure **191**. The RF coil array section **180** is a padded section (e.g., pillow-type structure). In certain embodiments, the RF coil array section **180** includes padding **192** disposed within the enclosure **191** as depicted in FIG. 3. In certain embodiments, the padding **192** is integrated as part of the enclosure **191** as depicted in FIG. 4. The RF coil array section **180** further includes a mechanically resistive structure **194** disposed within the enclosure **191**. The mechanically resistive structure **194** is configured to be manipulated (e.g., bent) to a particular position (e.g., bent position) and maintain the position without extraneous support (e.g., from a shim). Thus, the RF coil array section **180** is also configured to be manipulated (e.g., bent) to a particular position (e.g., bent position) and maintain the position without extraneous support. By manipulating the position of the RF coil array section **180**, the shape of the RF coil array section **180** is also changed. The mechanically resistive structure **194** is made of a material that is compatible with MRI. The RF coil **182** and its loops are flexible to accommodate the changing of the position (and shape) of the RF coil array section.

(34) FIGS. 5 and 6 are schematic diagrams of the RF coil array section **180** and its components for some of the sections forming a posterior RF receiving coil array assembly (e.g., posterior RF receiving coil array assembly **196** in FIG. 7). Each RF coil array section **180** includes the enclosure **191**. The RF coil **182** is disposed within the enclosure **191**. The RF coil array section **180** is a padded section (e.g., pillow-type structure). In certain embodiments, the RF coil array section **180** includes padding **192** disposed within the enclosure **191** as depicted in FIG. 3. In certain embodiments, the padding **192** is integrated as part of the enclosure **191** as depicted in FIG. 4. The RF coil array section **180** lacks a mechanically resistive structure and is not configured to be manipulated (e.g., bent) to a particular position (e.g., bent position). In certain embodiments, the RF coil **182** and its loops of the RF coil array sections **180** in FIGS. 5 and 6 may be rigid.

(35) FIG. 7 is a schematic diagram of a posterior RF receiving coil array assembly **196** disposed on the cradle **127** of the table **124** of the MRI system **100** in FIG. 1. As depicted, the posterior RF receiving coil array assembly **196** is disposed across an entirety of a top surface **197** of the cradle **127**. The posterior RF receiving coil array assembly **196** is coupled to (e.g., integrated with) the top surface **197** of the cradle **127**.

(36) The posterior RF receiving coil array assembly **196** is formed from multiple RF coil array sections **180**. As noted above, these sections **180** are padded and each section **180** includes the RF coil having loops. As depicted in FIG. 7, the posterior RF receiving coil array assembly **196** includes 9 RF coil array sections **180**. The number of RF coil array sections **180** in the posterior RF receiving coil array assembly **196** may vary.

(37) As depicted in FIG. 7, the RF coil array sections **180** located in different regions **198** (e.g., axial locations) along a longitudinal axis **200** of the posterior RF receiving coil array assembly **196** (and the cradle **127**). For example, a first plurality of RF coil array sections **180** are located at a first region **202** (e.g., first axial location), a second plurality of RF coil array sections **180** are located at a second region **204** (e.g., second axial location), and a third plurality of RF coil array sections **180** are located at a third region **206** (e.g., third axial location) along the longitudinal axis **200**. The number of regions **198** of multiple RF coil sections **180** at different axial locations along the longitudinal axis **200** may vary. As depicted, each region **198** includes 3 RF coil array sections **180**. In certain embodiments, each region **198** has the same number of RF coil array sections **180**.

In certain embodiments, at least two of the regions **198** may have a different number of RF coil array sections **180**. As depicted, at least two of the RF coil array sections **180** in each region **190** differ in dimensions. In certain embodiments, all of the RF coil array sections **180** in each region **190** may differ in dimensions. In certain embodiments, all of the RF coil array sections **180** in each region **198** may have the same dimensions. In certain embodiments, corresponding RF coil array sections **180** in each of the regions **198** may have different dimensions. In certain embodiments, corresponding RF coil array sections **180** in each of the regions **198** may have similar dimensions.

(38) Each region **198** is configured to be disposed on the top surface **197** of the cradle **127** below a posterior surface of a patient to be imaged. In each region **198**, at least one RF coil array section **180** is configured to be manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure (e.g., mechanically resistive structure **194** in FIGS. **3** and **4**). In particular, the at least one RF coil array section **180** is configured to be bent about the patient's anatomy from a posterior surface toward an anterior surface (and in some cases disposed about a portion of the anterior surface). In certain regions **198**, at least two RF coil array section **180** are configured to be manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure. In certain regions **198**, all of the RF coil array sections **180** are configured to be manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure.

(39) In certain regions, at least one RF coil array section **180** is configured to not be bent (i.e., lacks a mechanically resistive structure). As describe above, this type of section may include a rigid RF coil. In certain embodiments, the at least one RF coil array section **180** that is not configured to be bent is centrally located (e.g., RF coil array section **208** in FIG. **7**). In certain embodiments, the RF coil array section **180** (e.g., RF coil array section **208**) is flanked by RF coil array sections **180** that are configured to be bent (e.g., RF coil array sections **210**, **212** in the second region **204**).

(40) The cradle **127** includes lateral edges **214**, **216** and longitudinal edges or ends **218**, **220** relative to the longitudinal axis **200**. In certain embodiments, each of the RF coil array sections **180** (e.g., RF coil array sections **210**, **212**, **222**, **224**, **226**, **228**) adjacent the lateral edges **214**, **216** are configured to be manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure. In certain embodiments, some of the RF coil array sections **180** adjacent the longitudinal edges **218**, **220** are configured to be manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure. In certain embodiments, the centrally located RF coil array sections **180** adjacent the longitudinal edges **218**, **220** (e.g., RF coil array sections **230**, **232**) are not configured to be manipulated into a bent position (e.g., due to lack of a mechanically resistive structure). In certain embodiments, each RF coil array section **180** adjacent the longitudinal edges **218**, **220** (e.g., RF coil array sections **222**, **224**, **226**, **228**, **230**, **232**) are configured to be manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure.

(41) As mentioned above, the posterior RF receiving coil array assembly **196** is integrated on the top surface **197** on the cradle **127** of the table **124**. The posterior RF receiving coil array assembly **196** is integrated to the top surface **197** by coupling one or more of the RF coil array sections **180** to the top surface **197**. The one or more RF coil array sections **180** that are not configured to be bent (e.g., due to lacking a mechanically resistive structure) are the RF coil array sections **180** directly coupled to the top surface **197** via one or more fasteners (straps, hook and loop fasteners, rivets, etc.). For example, centrally located RF coil array section **208** (which is not configured to be manipulated into a bent position) may be directly coupled to top surface **197**. The one or more RF coil array sections **180** that are configured to be manipulated into a bent position are not directly coupled to the top surface **197** of the cradle **127** but are instead indirectly coupled to the top surface **197** (e.g., via the one or more RF coil array sections **180** that are not configured to be bent that are directly coupled to the top surface **197** of the cradle **127**).

(42) FIG. **8** is a schematic diagram of an end view of the posterior RF receiving coil array assembly

196 disposed within a bore **122** of the MRI scanner **102** of the MRI system **100**. As depicted in FIG. **8**, the posterior RF receiving coil array assembly **196** is integrated on the top surface **197** of the cradle **127** of the table **124**. As depicted, the posterior RF receiving coil array assembly includes multiple RF coil array sections **180**. In particular, the posterior RF receiving coil array assembly **196** includes a central RF coil array section **234** flanked by a pair of RF coil array sections **236**, **238**. Each RF coil array section **234**, **236**, **238** is a padded section. Each RF coil array section **234**, **236**, **238** is configured to be disposed beneath a posterior surface of a patient to be imaged. The central RF coil array section **234** is not configured to be manipulated or bent (i.e., the central RF coil array section **234** lacks a mechanically resistive structure). In certain embodiments, the central RF coil array section **234** includes a rigid RF coil. The RF coil array sections **236**, **238** each include a mechanically resistive structure that enables each RF coil array section **236**, **238** to be manipulated (e.g., bent) into a variety of positions (e.g., bent positions) and maintain these positions without extraneous support (e.g., without support from shims). Each RF coil array section **236**, **238** may have a flexible RF coil. As depicted in FIG. **8**, each RF coil array sections **236**, **238** can be manipulated in partially radial and partially circumferential direction as indicated by the arrows **240** into a plurality of different positions **242** (e.g., bent or curved positions) and maintain these positions **242** without extraneous support. The RF coil array sections **236**, **238** are configured to be bent about the patient's anatomy from a posterior surface toward an anterior surface (and in some cases disposed about a portion of the anterior surface).

(43) FIG. **9** is a schematic diagram of RF array coil sections **180** of the posterior RF receiving coil array assembly **196** disposed about a portion of the patient **126**. In FIG. **9**, the patient **126** may be an adult patient. As depicted, the posterior RF receiving coil array assembly **196** includes multiple RF coil array sections **180**. In particular, the posterior RF receiving coil array assembly **196** includes a central RF coil array section **244** flanked by a pair of RF coil array sections **246**, **248**. Each RF coil array section **244**, **246**, **248** is a padded section. Each RF coil array section **244**, **246**, **248** is configured to be disposed beneath a posterior surface **250** of the patient **126** to be imaged. The central RF coil array section **244** is not configured to be manipulated or bent (i.e., the central RF coil array section **244** lacks a mechanically resistive structure). In certain embodiments, the central RF coil array section **244** includes a rigid RF coil. The RF coil array sections **246**, **248** each include a mechanically resistive structure that enables each RF coil array section **246**, **248** to be manipulated (e.g., bent) into a variety of positions (e.g., bent positions) and maintain these positions without extraneous support (e.g., without support from shims). Each RF coil array section **246**, **248** may have a flexible RF coil. As depicted in FIG. **9**, each RF coil array sections **246**, **248** bent about the patient **126** from the posterior surface **250** toward an anterior surface **252**. As depicted in FIG. **9**, a separate surface coil array **254** (e.g., anterior RF receiving coil array) is during the MRI imaging of the patient **126**.

(44) FIG. **10** is a schematic diagram of a top view of the posterior RF receiving coil array assembly **196** being utilized with a pediatric patient **126**. The posterior RF receiving coil array assembly **196** is similar to that described in FIG. **7**. In particular, the posterior RF receiving coil array assembly **196** disposed on the cradle **127** of the table **124** of the MRI system **100** in FIG. **1**. As depicted, the posterior RF receiving coil array assembly **196** is disposed across the entirety of the top surface **197** of the cradle **127**. The posterior RF receiving coil array assembly **196** is coupled to (e.g., integrated with) the top surface **197** of the cradle **127**.

(45) The posterior RF receiving coil array assembly **196** is formed from multiple RF coil array sections **180**. As noted above, these sections **180** are padded and each section **180** includes the RF coil having loops. As depicted in FIG. **7**, the RF coil array sections **180** located in different regions **198** (e.g., axial locations) along the longitudinal axis **200** of the posterior RF receiving coil array assembly **196** (and the cradle **127**). For example, a first plurality of RF coil array sections **180** are located at the first region **202** (e.g., first axial location), a second plurality of RF coil array sections **180** are located at the second region **204** (e.g., second axial location), and a third plurality of RF

coil array sections **180** are located at the third region **206** (e.g., third axial location) along the longitudinal axis **200**.

(46) Each region **198** is configured to be disposed on the top surface **197** of the cradle **127** below a posterior surface of the pediatric patient **126** to be imaged. As depicted, the pediatric patient **126** is centrally located in the region **204**. RF coil array sections **180** flanking the pediatric patient **126** are manipulated (e.g., bent) into a particular position (e.g., bent position) due to having a mechanically resistive structure (e.g., mechanically resistive structure **194** in FIGS. **3** and **4**). In particular, these RF coil array sections **180** are bent about the pediatric patient **126** from the posterior surface toward (and disposed about) the anterior surface. Since the RF coil array sections **180** bend about both the posterior and anterior surface of the pediatric patient **126**, no additional surface coils (e.g., anterior RF receiving coil array) are needed to image the pediatric patient **126**.

(47) FIG. **11** is a schematic diagram of RF array coil sections **180** of the posterior RF receiving coil array assembly **196** disposed about a portion of the patient **126**. In FIG. **11**, the patient **126** may be a pediatric patient. As depicted, the posterior RF receiving coil array assembly **196** includes multiple RF coil array sections **180**. In particular, the posterior RF receiving coil array assembly **196** includes a central RF coil array section **256** flanked by a pair of RF coil array sections **258**, **260**. Each RF coil array section **256**, **258**, **260** is a padded section. Each RF coil array section **256**, **258**, **260** is configured to be disposed beneath the posterior surface **250** of the patient **126** to be imaged. The central RF coil array section **256** is not configured to be manipulated or bent (i.e., the central RF coil array section **256** lacks a mechanically resistive structure). In certain embodiments, the central RF coil array section **256** includes a rigid RF coil. The RF coil array sections **258**, **260** each include a mechanically resistive structure that enables each RF coil array section **258**, **260** to be manipulated (e.g., bent) into a variety of positions (e.g., bent positions) and maintain these positions without extraneous support (e.g., without support from shims). Each RF coil array section **258**, **260** may have a flexible RF coil. As depicted in FIG. **11**, each RF coil array sections **258**, **260** bent about the patient **126** from the posterior surface **250** toward an anterior surface **252** and then across the anterior surface **252**. As depicted in FIG. **11**, a separate surface coil array **254** (e.g., anterior RF receiving coil array) is during the MRI imaging of the patient **126**. Since the RF coil array sections **258**, **260** bend about both the posterior surface **250** and the anterior surface **252** of the pediatric patient **126**, no additional surface coils (e.g., anterior RF receiving coil array) are needed to image the pediatric patient **126**.

(48) FIG. **12** is a schematic diagram of an RF coil array section **180** of the posterior RF receiving coil array assembly **196** being positioned to lift a portion of the patient (e.g., head **262**). The patient **126** is disposed across the posterior RF receiving coil array assembly **196** (e.g., as described in FIG. **7**). Depending on the orientation of the patient **126** when moved into the bore of the MRI scanner (e.g., head first or feet first), one of the RF coil array sections **180** (e.g., RF coil array section **230** or **232**) at the longitudinal ends **218**, **220** may be manipulated (e.g., bent) to lift or tilt the head **262** of the patient **126** and maintain the positioning of the head **262** without any extraneous support (e.g., without a shim).

(49) FIG. **13** is a schematic diagram of an RF coil array section **180** of the posterior RF receiving coil array assembly **196** being positioned to lift a portion of the patient (e.g., legs **264** and feet **266**). The patient **126** is disposed across the posterior RF receiving coil array assembly **196** (e.g., as described in FIG. **7**). Depending on the orientation of the patient **126** when moved into the bore of the MRI scanner (e.g., head first or feet first), one of the RF coil array sections **180** (e.g., RF coil array section **230** or **232**) at the longitudinal ends **218**, **220** may be manipulated (e.g., bent) to lift or tilt the legs **264** and the feet **266** of the patient **126** and maintain the positioning of the legs **264** and the feet **266** without any extraneous support (e.g., without a shim).

(50) Technical effects of the disclosed subject matter include enabling the posterior RF receiving coil array to be disposed closer to the anatomy of the patient being imaged. The disclosed embodiments enable for better in-table patient positioning and remove the need for comfort tilt

accessories. In addition, the disclosed embodiments alleviate issues of carrying around large and somewhat clumsy surface coils for quicker workflow. The disclosed embodiments further provide better contour fitting and comfort to different patient types which boosts image quality. The disclosed embodiments reduce trips to coils closet and overall setup time for the patients while also providing a closer loop-to-anatomy distance to optimize image quality. Even further, by improving workflow with less trips and providing more integrated capability all at the table, the throughput of scanning patients is increased.

(51) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

(52) This written description uses examples to disclose the present subject matter, including the best mode, and also to enable any person skilled in the art to practice the subject matter, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A magnetic resonance imaging (MRI) system, comprising: an MRI scanner having a bore; and a table comprising a cradle and a radio frequency (RF) receiving coil array assembly integrated on a top surface of the cradle, wherein the RF receiving coil array assembly extends across an entirety of the top surface of the cradle, wherein the table is configured to move a subject to be imaged disposed on the RF receiving coil array assembly into and out of the bore of the MRI scanner, wherein the RF receiving coil array assembly comprises a central section comprising a rigid RF coil enclosed within an enclosure, wherein the central section is not configured to be bent, and wherein the RF receiving coil array assembly comprises a plurality of sections that are physically separate from each other and the central section, each section of the plurality of sections comprises a padding and a flexible RF coil comprising a plurality of loops enclosed within a flexible enclosure, and each section of the plurality of sections is configured to be manipulated into a bent position, wherein each section of the plurality of sections is configured to be manipulated into a plurality of different bent positions, wherein the plurality of sections comprises a first section adjacent a first longitudinal end of the central section, a second section adjacent to a second longitudinal end of the central section opposite the first longitudinal end, a third section adjacent to a first side of the central section, a fourth section adjacent to a second side of the central section opposite the first side, the first side and the second side extending between the first longitudinal end and the second longitudinal end, a fifth section and a sixth section flanking the first section, and a seventh section and an eighth section flanking the second section.

2. The MRI system of claim 1, wherein at least one section of the plurality of sections is located adjacent an edge of the cradle.

3. The MRI system of claim 1, wherein at least one section of the plurality of sections is configured to be bent about the subject from a posterior surface of the subject toward an anterior surface of the subject.

4. The MRI system of claim 3, wherein the at least one section of the plurality of sections in the

bent position is configured to be disposed about at least a portion of the anterior surface of the subject.

5. The MRI system of claim 1, wherein the RF receiving coil array assembly is configured to be utilized on both a posterior surface and an anterior surface of a pediatric subject without utilizing any additional RF receiving coil array assembly during an imaging scan of the pediatric subject by the MRI system.

6. A radio frequency (RF) receiving coil array assembly for a magnetic resonance imaging (MRI) system, comprising: a central section comprising a rigid RF coil enclosed within an enclosure, wherein the central section is not configured to be bent; and a plurality of sections that are physically separate from each other and the central section, wherein each section of the plurality of sections comprises a padding and a flexible RF coil comprising a plurality of loops enclosed within a flexible enclosure, and each section of the plurality of sections is configured to be manipulated into a bent position, wherein each section of the plurality of sections is configured to be manipulated into a plurality of different bent positions, wherein the plurality of sections comprises a first section adjacent a first longitudinal end of the central section, a second section adjacent to a second longitudinal end of the central section opposite the first longitudinal end, a third section adjacent to a first side of the central section, a fourth section adjacent to a second side of the central section opposite the first side, the first side and the second side extending between the first longitudinal end and the second longitudinal end, a fifth section and a sixth section flanking the first section, and a seventh section and an eighth section flanking the second section, wherein the RF receiving coil array assembly is configured to be integrated on a top surface of a cradle of a table configured to move a subject to be imaged disposed on the RF receiving coil array assembly into and out of a bore of an MRI scanner of the MRI system, and wherein the RF receiving coil array assembly is configured to extend across an entirety of the top surface of the cradle.

7. The RF receiving coil array assembly of claim 6, wherein at least one section of the plurality of sections is configured to be bent about the subject from a posterior surface of the subject toward an anterior surface of the subject.

8. The RF receiving coil array assembly of claim 6, wherein at least one section of the plurality of sections is located adjacent an edge of the cradle.
